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Prepayment Tariff

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1. Introduction

The Power & Electricity Department (P&ED), Government of Mizoram, is the licensed electricity distribution utility responsible for the supply of electricity to consumers across the State of Mizoram and for billing and revenue collection in accordance with the tariff orders issued by the Joint Electricity Regulatory Commission (JERC) for Manipur & Mizoram. P&ED operates and maintains the distribution network at both Low Tension (LT) and High Tension (HT) levels and serves a diverse mix of consumer categories, which are classified based on the nature of use, voltage of supply, and consumption characteristics.

The consumer categories served by P&ED, as per the prevailing tariff framework, are broadly classified as follows:

Low Tension (LT) Categories

- Kutir Jyoti (lifeline domestic households)
- Domestic Service
- Non-Profit Public Service (earlier Non-Domestic)
- Commercial Service
- Public Lighting (street and thoroughfare lighting)
- Public Water Works
- Irrigation & Agriculture Service
- LT Industrial Service
- Electric Vehicle Charging Stations (LT)

High Tension (HT) Categories

- HT Domestic Service
- HT Non-Profit Public Service
- HT Commercial Service
- HT Public Water Works
- HT Irrigation & Agriculture Service
- HT Industrial Service (including Time-of-Day option where applicable)
- Bulk Supply within the State
- Electric Vehicle Charging Stations (HT)

Each of the above categories is governed by a distinct tariff structure, with LT consumers billed primarily on the basis of contracted load in kW and energy consumption in kWh, while HT consumers are billed based on contracted or billing demand in kVA and energy consumption in kVAh, along with applicable demand charges, Time-of-Day rates, and subsidy provisions were notified.

2. Rationale for Transition from Postpaid to Prepaid Billing

Under the existing postpaid billing system, electricity is supplied throughout the billing cycle and charges are recovered after consumption based on periodic meter readings and bill generation. While this mechanism complies with the approved tariff framework, it inherently results in a time gap between energy supply and revenue realisation, leading to higher outstanding receivables, delayed collections, and increased administrative effort in billing, collection, and enforcement.

The transition to prepaid metering represents a change in the mode of payment and recovery, not a change in tariff. All JERC-approved tariff rates, slabs, demand charges, and subsidy mechanisms continue to apply in full. Prepaid metering enables advance payment and real-time or near-real-time deduction of charges based on actual consumption, offering the following advantages:

- **For P&ED (Utility)**

- Improved and predictable cash flow with revenue realisation closer to the time of supply
- Reduction in arrears, bad debts, and collection inefficiencies
- Lower operational costs related to billing, disconnection, and reconnection
- Enhanced control over revenue leakage and billing disputes
- Availability of granular consumption data for system planning and loss reduction

- **For Consumers**

- Greater transparency and visibility of electricity usage and charges
- Improved control over consumption and expenditure
- Reduced risk of bill shocks and disputes
- Timely alerts and balance information supporting informed usage decisions

The proposed prepaid framework therefore strengthens financial sustainability for the utility while improving transparency, efficiency, and service quality for consumers, without altering the underlying tariff structure approved by the Commission.

3. Consumer Profile

The consumer profile of P&ED, Mizoram for FY 2022–23 reflects a predominance of Low Tension (LT) consumers in terms of numbers, particularly under the Domestic and Kutir Jyoti categories, while High Tension (HT) consumers, though fewer in number, contribute a significant share of total energy sales due to higher connected loads and consumption intensity.

Domestic LT consumers form the largest consumer base, accounting for the majority of connections and energy sales among LT categories. Public Water Works, Commercial, and Industrial consumers contribute materially to overall demand, especially under HT supply. Bulk Supply and Outside State sale, despite limited consumer counts, account for a sizable portion of energy sales.

This distribution highlights the diverse nature of P&ED's consumer mix and underlines the importance of differentiated tariff structures and billing mechanisms, including prepaid metering, tailored to consumption patterns across categories.

Table: Consumer Profile and Energy Sales for FY 2022–23 (Audited Actuals)

Sl. No.	Category	Approx No. of Consumers
1(a)	Kutir Jyoti – LT	17,441
1(b)	Domestic – LT	243,058
1(c)	Domestic – HT	11
2(a)	Non-Profit Public Purpose – LT	8,212
2(b)	Non-Profit Public Purpose – HT	55
3(a)	Commercial – LT	13,176
3(b)	Commercial – HT	497
4	Public Lighting – LT	520
5(a)	Irrigation & Agriculture – LT	52
5(b)	Irrigation & Agriculture – HT	23
6(a)	Public Water Works – LT	177
6(b)	Public Water Works – HT	55
7(a)	Industrial – LT	1,027
7(b)	Industrial – HT	26
8	Bulk Supply – HT	146
9	Temporary Connection & Theft	13
10	Outside State Sale – UI / Trading	–
	Grand Total	284,489

4. Demand

The allocation of power to the State of Mizoram comprises firm and un-firm (un-allocated) power from various Central Generating Stations, NEC-funded Baramura Gas Based Thermal Power Plant, and other approved sources.

During FY 2022-23, the total allocated capacity available to the Power & Electricity Department (P&ED), Mizoram was 272.02 MW. The peak load demand of the State during FY 2022-23 was recorded at 159.12 MW on 19.01.2023.

In FY 2023-24 (provisional), the peak load demand increased to 163.10 MW, recorded on 27.01.2024, reflecting a continued growth in electricity demand across domestic, commercial, and public service sectors.

5. Tariff Category Configuration

Detailed tariff-wise information for each Low Tension (LT) and High Tension (HT) consumer category has been reviewed and documented as part of this submission for approval from PED Mizoram.

For every tariff category, the corresponding billing determinants, including fixed or demand charges, energy charges, slab structures, Time-of-Day applicability, subsidy treatment, and pro-rata rules, have been mapped and configured in the proposed prepaid metering and billing engine.

The prepaid configuration mirrors the existing postpaid tariff framework approved by the Joint Electricity Regulatory Commission (JERC) for Manipur & Mizoram and does not alter any tariff rates, category definitions, or subsidy mechanisms.

Our understanding and implementation of prepaid calculations for all LT and HT categories are fully aligned with the prevailing tariff orders, Supply Code provisions, and regulatory principles applicable to P&ED Mizoram, ensuring regulatory compliance, billing accuracy, and transparency for all consumer categories.

6. Low Tension (LT) Categories

Low Tension (LT) categories cover consumers supplied at LT voltage levels and are primarily billed based on contracted load in kilowatt (kW) and energy consumption in kilowatt-hour (kWh).

These categories include residential, public service, commercial, industrial, and other utility-related consumers with relatively lower connected loads compared to HT consumers. For prepaid implementation, the approved LT tariff structures, including fixed charges, energy charges, slab-wise rates, subsidies, and pro-rata billing rules,

have been configured without any change to the tariff design, ensuring full alignment with P&ED Mizoram and JERC-approved norms.

6.1 LT Category - 1: Kutir Jyoti Service

6.1.1 Applicability:

This category is applicable to economically weaker section households provided with basic electricity supply for domestic use, such as lighting and limited household appliances, under the Kutir Jyoti scheme. The supply is intended as a lifeline service for low-income consumers and is subject to specific consumption and load limits.

6.1.2 Rates:

Component	Rates
Fixed Charge	₹25.00 per month per connection
Energy Charge – First 20 kWh	₹2.55 per kWh
Energy Charge – Above 20 kWh	₹3.55 per kWh

Note: If the total consumption in any consecutive three months exceeds 45 kWh, the connection should be converted to LT Category-2 (Domestic).

6.1.3 Pre-Payment Configuration

(a) Parameterization

As the prepaid metering engine is tightly integrated with the ZONOS Meter Data Management (MDM) system, all consumer and device-related master data are received through a synchronized master data interface using a unique device ID and consumer ID.

The following parameters are validated and used for the initial tariff configuration and prepaid billing setup at the time of service activation or migration to prepaid mode:

- **Contracted Load (kW):** Used for applying fixed charges and validating tariff eligibility.
- **Installation Date (dd-mm-yyyy):** Considered as the effective start date for billing and pro-rata calculations.
- **Billing Days:** Calculated as [Last day of the current month - Installation Date] and used for determining partial-month billing.
- **Billing Factor:** Derived as [Total number of days in the month ÷ Billing Days] and applied to normalize fixed and energy charges for the billing period.
- **Opening Balance (₹):** The initial prepaid balance available in the consumer's account at the time of activation, from which energy and fixed charge deductions commence.

These parameters ensure accurate tariff application, correct pro-rata billing, and seamless alignment between the prepaid engine, MDM system, and the approved tariff framework of P&ED Mizoram.

Consumer Category: Kutir Jyoti (LT): Sample Data		
Contracted Load	0.5	kW
Installation Date	01-12-2025	Date
Current Month days	31	Days
Last Date of Month	31-12-2025	Date
Bill Days	31	[Last day of the current month - Installation Date]
Bill factor	1.00	[Total number of days in the month ÷ Billing Days]
Opening Balance	₹ 20	Initial prepaid balance

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate as approved in the tariff order. For prepaid billing, the fixed charge is applied on a pro-rata basis for the relevant billing period, especially for partial months, and deducted from the prepaid balance without any modification to the tariff.

$$\text{Fixed Charge} = \text{Contracted Load} \times \text{Fixed Charge Rate} \times (\text{Billing Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated based on the actual energy consumption recorded by the meter and the applicable energy rate as per the approved tariff structure. In the prepaid system, energy charges are deducted in near real time or at defined intervals from the prepaid balance, strictly following slab-wise or flat energy rates without any alteration.

$$\text{Energy Charge} = \text{Energy Consumption (kWh)} \times \text{Applicable Energy Rate}$$

For Kutir Jyoti consumers, consumption up to the first 20 kWh in a billing period is charged at a lower rate, while any consumption beyond this limit is charged at the higher slab rate, with deductions made from the prepaid balance accordingly.

$$\begin{aligned} \text{Energy Charge} = & (\text{Min}(\text{Total kWh}, 20) \times 2.55) \\ & + (\text{Max}(\text{Total kWh} - 20, 0) \times 3.55) \end{aligned}$$

d) Surcharge Calculation

In the prepaid system, the surcharge amount is computed after the base charges and deducted from the consumer's prepaid balance as part of the total payable amount.

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times \text{Applicable Surcharge Rate}$$

For PED Mizoram, a surcharge at the rate of 2% is applied on the total of energy charges and fixed charges, as per the applicable tariff provisions. In the prepaid system, this surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction View:

In the prepaid system, daily electricity consumption is recorded from the meter and energy charges are calculated based on applicable tariff rates. Fixed charges are apportioned on a daily basis, and applicable surcharge is computed on the sum of energy and fixed charges. The total daily deduction is then debited from the opening balance to arrive at the current available balance for the consumer

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



2.1 Kutir Jyoti (LT)

6.2 LT Category - 2: Domestic Service

6.2.1 Applicability:

This category is applicable for electricity supplied to residential premises for domestic use such as lighting, fans, heating, refrigeration, pumping and other

household appliances. It covers supply used purely for household purposes and excludes usage of a commercial nature.

6.2.2 Rates:

Component	Rates
Fixed Charge	₹50 per kW per month
Energy Charge – First 100 kWh	₹4.90 per kWh
Energy Charge – Next 100 kWh (101–200 kWh)	₹7.10 per kWh
Energy Charge – Above 200 kWh	₹8.20 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.2.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS MDM system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totalnumberofdaysinthemonth \div BillDays$
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the Domestic LT category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period, particularly in cases of new connections or mid-month installations and is deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

For LT Category–2 (Domestic Service), energy charges are levied on a slab-wise basis to encourage efficient consumption. Consumption up to the initial slab is charged at a lower rate, with higher rates applicable as consumption increases in subsequent slabs. The prepaid engine maintains a cumulative monthly unit counter to ensure correct slab allocation as consumption progresses through the billing cycle. The applicable slabs are as follows:

- First 100 kWh of monthly consumption are charged at ₹4.90 per kWh

- Next 100 kWh (101–200 kWh) are charged at ₹7.10 per kWh
- Consumption above 200 kWh is charged at ₹8.20 per kWh

Energy charges are calculated based on actual meter readings and deducted from the prepaid balance strictly in accordance with these slab rates, without any alteration to the approved tariff.

$$\begin{aligned} \text{Energy Charge} = & (\text{MIN} (\text{Total kWh}, 100) \times 4.90) \\ & + (\text{MIN} (\text{MAX} (\text{Total kWh} - 100, 0), 100) \times 7.10) \\ & + (\text{MAX} (\text{Total kWh} - 200, 0) \times 8.20) \end{aligned}$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable Domestic LT tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed based on actual daily consumption with cumulative slab tracking for the billing period, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\begin{aligned} \text{Gross Daily Deduction} &= \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge} \\ \text{Current Balance} &= \text{Previous Balance} - \text{Gross Daily Deduction} \end{aligned}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



6.3 LT Category - 3: Non-Profit Public Service

6.3.1 Applicability

This category is applicable for electricity supply to non-profit institutions and public service establishments such as educational institutions, government offices, religious places, charitable institutions, community halls, and similar premises. Supply under this category is strictly meant for public or community service purposes and excludes usage of a commercial or profit-oriented nature.

6.3.2 Rates

Component	Rate
Fixed Charge	₹60.00 per month per kW of contracted load
Energy Charge – First 150 kWh	₹7.60 per kWh
Energy Charge – All units above 150 kWh	₹8.30 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.3.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = $\text{Lastdayofcurrentmonth} - \text{InstallationDate}$
- Bill Factor = $\text{Totalnumberofdaysinthemonth} \div \text{BillDays}$
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the LT Non-Profit Public Service category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the

billing period, particularly in cases of new connections or mid-month installations and is deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges for LT Non-Profit Public Service are applied on a slab-wise basis. Consumption up to the first slab is charged at a lower rate, while consumption beyond the slab threshold is charged at a higher rate. The prepaid engine maintains a cumulative monthly unit counter to ensure correct slab allocation during the billing cycle.

Applicable slabs:

- First 150 kWh at ₹7.60 per kWh
- All units above 150 kWh at ₹8.30 per kWh

Energy charges are computed strictly as per these slabs and deducted from the prepaid balance without any modification to the approved tariff

$$\begin{aligned} \text{Energy Charge} = & (\text{MIN}(\text{Total kWh}, 150) \times 7.60) \\ & + (\text{MAX}(\text{Total kWh} - 150, 0) \times 8.30) \end{aligned}$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

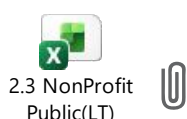
In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT Non-Profit Public Service tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\begin{aligned} \text{Gross Daily Deduction} &= \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge} \\ \text{Current Balance} &= \text{Previous Balance} - \text{Gross Daily Deduction} \end{aligned}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



6.4 LT Category - 4: Commercial Service

6.4.1 Applicability

This category is applicable for electricity supply to premises used for commercial activities such as shops, offices, hotels, restaurants, business establishments, private service providers, and similar premises where electricity is used for profit-oriented or commercial purposes. Supply under this category excludes domestic and non-profit public service usage.

6.4.2 Rates

Component	Rate
Fixed Charge	₹80.00 per month per kW of contracted load
Energy Charge – First 150 kWh	₹8.20 per kWh
Energy Charge – All units above 150 kWh	₹8.45 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.4.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = *Lastdayofcurrentmonth – InstallationDate*
- Bill Factor = *Totalnumberofdaysinthemonth ÷ BillDays*

- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the LT Commercial Service category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period, particularly in cases of new connections or mid-month installations and is deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges for LT Commercial Service are applied on a slab-wise basis. Consumption up to the first slab is charged at a lower rate, while consumption beyond the slab threshold is charged at a higher rate. The prepaid engine maintains a cumulative monthly unit counter to ensure correct slab allocation during the billing cycle.

Applicable slabs:

- First 150 kWh at ₹8.20 per kWh
- All units above 150 kWh at ₹8.45 per kWh

Energy charges are computed strictly as per these slabs and deducted from the prepaid balance without any modification to the approved tariff.

$$\begin{aligned} \text{Energy Charge} = & (\text{MIN}(\text{Total kWh}, 150) \times 8.20) \\ & + (\text{MAX}(\text{Total kWh} - 150, 0) \times 8.45) \end{aligned}$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT Commercial Service tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

Gross Daily Deduction = Daily Fixed Charge + Daily Energy Charges + Daily Surcharge

Current Balance = Previous Balance - Gross Daily Deduction

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



2.4 Commercial(LT)

6.5 LT Category - 5: Public Lighting Service

6.5.1 Applicability

This category is applicable for electricity supply used exclusively for public street lighting, road lighting, and thoroughfare lighting provided by local bodies, municipal authorities, government departments, or other authorised agencies. The supply under this category is restricted to lighting of public roads, streets, and public spaces and does not include any commercial or private usage.

6.5.2 Rates

Component	Rate
Fixed Charge	₹80 per kW per month
Energy Charge	₹11.35 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.5.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totalnumberofdaysinthemonth \div BillDays$
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the LT Public Lighting Service category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period, particularly in cases of new connections or mid-month installations and is deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated based on the actual energy consumption recorded by the meter and are applied at a flat rate as approved for the LT Public Lighting Service category. The prepaid engine computes energy charges on actual consumption and deducts the corresponding amount from the prepaid balance without any alteration to the tariff structure.

Energy Rate: ₹11.35 per kWh

$$\text{Energy Charge} = \text{Total Energy Consumption (kWh)} \times 11.35$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT Public Lighting Service tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\text{Gross Daily Deduction} = \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge}$$

$$\text{Current Balance} = \text{Previous Balance} - \text{Gross Daily Deduction}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



2.5 Public Light(LT)

6.6 LT Category - 6: Public Water Works

6.6.1 Applicability

This category is applicable for electricity supply used for operation of public water supply systems, including water pumping stations, treatment plants, and related infrastructure operated by government departments, local bodies, municipal authorities, or other authorised agencies. Supply under this category is strictly limited to activities related to public water supply and excludes commercial or industrial usage.

6.6.2 Rates

Component	Rate
Fixed Charge	₹90 per kW per month
Energy Charge	₹11.10 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.6.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totalnumberofdaysinthemonth \div BillDays$
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the LT Public Water Works category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period, particularly in cases of new connections or mid-month installations, and is deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated based on the actual energy consumption recorded by the meter and are initially computed at the applicable full cost rate. Subsidy, where applicable, is adjusted to arrive at the net payable energy charge as approved for the LT Public Water Works category. The prepaid engine deducts the net payable energy charges from the prepaid balance.

Net Energy Rate: ₹11.10 per kWh

$$\text{Energy Charge} = \text{Total Energy Consumption (kWh)} \times 11.10$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT Public Water Works tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\text{Gross Daily Deduction} = \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge}$$

$$\text{Current Balance} = \text{Previous Balance} - \text{Gross Daily Deduction}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



2.6 Public Water(LT)

6.7 LT Category - 7: Irrigation & Agriculture Service

6.7.1 Applicability

This category is applicable for electricity supply used for agricultural and irrigation purposes, including operation of irrigation pumps, lift irrigation systems, and other agricultural activities carried out by individual farmers, farmer groups, cooperatives, or government-supported irrigation schemes. Supply under this category is restricted to agricultural use and excludes any commercial or industrial activity.

6.7.2 Rates

Component	Rate
Fixed Charge	₹50 per kW per month
Energy Charge	₹3.80 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.7.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = *Lastdayofcurrentmonth – InstallationDate*
- Bill Factor = *Totalnumberofdaysinthemonth ÷ BillDays*
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the LT Irrigation & Agriculture Service category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period, particularly in cases of new connections or mid-month installations, and is deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated based on the actual energy consumption recorded by the meter and are applied at a flat rate as approved for the LT Irrigation & Agriculture Service category. The prepaid engine computes energy charges on actual consumption and deducts the corresponding amount from the prepaid balance without any alteration to the tariff structure.

Energy Rate: ₹3.80 per kWh

$$\text{Energy Charge} = \text{Total Energy Consumption (kWh)} \times 3.80$$

d) Surcharge Calculation

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As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT Irrigation & Agriculture Service tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\text{Gross Daily Deduction} = \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge}$$

$$\text{Current Balance} = \text{Previous Balance} - \text{Gross Daily Deduction}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



2.7 Irrigation(LT)

6.8 LT Category - 8: Industrial Service

6.8.1 Applicability

This category is applicable for electricity supply used for industrial activities such as manufacturing, processing, fabrication, preservation, or production of goods. It includes small and medium industrial units, workshops, mills, agro-processing units,

cold storage, and similar establishments operating at Low Tension (LT) voltage and not covered under HT Industrial Service.

6.8.2 Rates

Component	Rate
Fixed Charge	₹80.00 per month per kW of contracted load
Energy Charge – First 400 kWh	₹7.10 per kWh
Energy Charge – All units above 400 kWh	₹8.05 per kWh
Surcharge	2% on (Energy Charges + Fixed Charge)

6.8.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = $\text{Lastdayofcurrentmonth} - \text{InstallationDate}$
- Bill Factor = $\text{Totalnumberofdaysinthemonth} \div \text{BillDays}$
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load of the consumer and the applicable fixed charge rate approved for the LT Industrial Service category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period and deducted from the prepaid balance as per the configured deduction cycle.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges for LT Industrial Service are applied on a slab-wise basis. Consumption up to the first slab is charged at a lower rate, while consumption beyond the slab threshold is charged at a higher rate. The prepaid engine maintains a cumulative monthly unit counter to ensure correct slab allocation throughout the billing cycle.

The applicable slabs are:

- First 400 kWh of monthly consumption at ₹7.10 per kWh
- All units above 400 kWh at ₹8.05 per kWh

Energy charges are computed strictly as per these slabs and deducted from the prepaid balance without any modification to the approved tariff.

$$\text{Energy Charge} = (\text{MIN}(\text{Total kWh}, 400) \times 7.10) \\ + (\text{MAX}(\text{Total kWh} - 400, 0) \times 8.05)$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT Industrial Service tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption with cumulative slab tracking, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\text{Gross Daily Deduction} = \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge}$$

$$\text{Current Balance} = \text{Previous Balance} - \text{Gross Daily Deduction}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



6.9 LT Category - 9: Electric Vehicle Charging Stations

6.9.1 Applicability

This category is applicable for electricity supply used exclusively for charging of electric vehicles through authorised charging infrastructure installed at public or private locations. It covers EV charging stations operated by individuals, commercial entities, government agencies, or other authorised service providers, and excludes any non-EV related usage.

6.9.2 Rates

Component	Rate
Fixed Charge	₹75.00 per month per kW of contracted load
Energy Charge	₹8.20 per kWh (for all units)
Surcharge	2% on (Energy Charges + Fixed Charge)

6.9.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

The prepayment engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer and meter master data are synchronized using a unique Consumer ID and Device ID. The following parameters are received and validated for initial tariff configuration and prepaid billing setup:

- Contracted Load (kW)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totalnumberofdaysinthemonth \div BillDays$
- Opening Balance (₹)

b) Fixed Charge Calculation

The fixed charge is calculated based on the contracted load and the applicable fixed charge rate approved for the LT Electric Vehicle Charging Station category. For prepaid billing, the fixed charge is applied on a pro-rata basis for the billing period and deducted from the prepaid balance.

$$\text{Fixed Charge} = \text{Contracted Load (kW)} \times \text{Fixed Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges for LT Electric Vehicle Charging Stations are applied at a single flat rate for all units consumed, without any slab differentiation. The prepaid engine computes energy charges directly on actual consumption and deducts the amount from the prepaid balance.

Energy Rate: ₹8.20 per kWh

$$\text{Energy Charge} = \text{Total Energy Consumption (kWh)} \times 8.20$$

d) Surcharge Calculation

As per PED Mizoram provisions, a surcharge is applicable on the total of energy charges and fixed charges. In the prepaid system, the surcharge is calculated after determining the base charges and is deducted from the consumer's prepaid balance along with other applicable charges.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Energy Charge} + \text{Fixed Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update (*Revised logic applied*)

In the prepaid system, electricity consumption is recorded on a daily basis and priced using the applicable LT EV Charging tariff logic. The fixed charge is apportioned on a daily pro-rata basis, energy charges are computed on actual daily consumption at the flat rate, and the applicable surcharge is calculated on the sum of fixed and energy charges. The resulting gross daily deduction is debited from the consumer's prepaid balance to arrive at the updated current balance.

$$\text{Gross Daily Deduction} = \text{Daily Fixed Charge} + \text{Daily Energy Charges} + \text{Daily Surcharge}$$

$$\text{Current Balance} = \text{Previous Balance} - \text{Gross Daily Deduction}$$

Day	Units Consumed (kWh)	Fixed Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note: Fixed charge shown is a daily pro-rata amount. Energy charges follow applicable slab rates. Surcharge is calculated at 2% on the sum of fixed and energy charges. Balances will be rounded for illustration.

f) Link to the Sample Calculation



7. High Tension (HT) Categories

High Tension (HT) tariff categories apply to consumers with higher demand and energy requirements, where electricity supply is provided at HT voltage levels. Billing under HT categories is based on contract demand in kVA and energy consumption in kVAh, reflecting both active and reactive power usage. The HT prepaid framework ensures accurate, transparent, and real-time billing through daily provisional deductions and month-end reconciliation, fully aligned with PED Mizoram and JERC tariff principles.

The tariffs are applicable for Consumer availing supply at voltage above 400Volts irrespective of connected load/contracted demand. It is mandatory to supply with voltage above 400V, to consumer having a contracted Load of above 50 kW or Contracted Demand of above 55.56 kVA, as per clause 3.2 of JERC for M&M (Electricity Supply Code) Regulations, 2013, with upto date amendment.

7.1 HT Category - 1: Domestic Service

7.1.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage for domestic/residential use, for the same purposes covered under LT Category–2 (Domestic Service). It includes residential premises with higher demand requirements where billing is based on contract demand (kVA) and energy consumption in kVAh as recorded by the HT meter.

7.1.2 Rates

Component	Rate
Demand Charge	₹50.00 per month per kVA of Billing Demand
Energy Charge	₹8.65 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.1.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system, through which consumer, meter, and electrical

parameters are synchronised using a unique **Consumer ID** and **Device ID**. In addition to the standard master data used for LT consumers, HT billing requires certain electrical parameters to ensure accurate demand and energy computation. The following parameters are received, validated, and used for initial tariff configuration and prepaid billing setup:

- **Contract Demand (kVA):** Used for determination of billing demand and demand charge calculation.
- **Installation Date (dd-mm-yyyy):** Treated as the effective start date for billing and pro-rata calculations.
- **Bill Days:** $\text{Last day of the current month} - \text{Installation Date}$.
- **Bill Factor:** $\text{Total number of days in the month} \div \text{Bill Days}$
- **Power Factor (PF):** Used to convert active energy (kWh) measured by the meter into apparent energy (kVAh) for billing purposes.
- **Multiplying Factor (MF):** Applied to scale meter-recorded values to actual system values based on CT/PT ratios.
- **Opening Balance (₹):** Initial prepaid balance available in the consumer's account at the time of activation.

Power Factor (PF):

HT tariffs are billed on kVA and kVAh to account for both active and reactive components of power. Since meters may internally record energy in kWh, the Power Factor is used to convert kWh to kVAh so that billing reflects the actual apparent energy drawn by the consumer and aligns with the approved HT tariff design.

$$\text{kVAh} = \text{kWh} \div \text{Power Factor}$$

Multiplying Factor (MF):

HT meters are connected through Current Transformers (CTs) and Potential Transformers (PTs). The meter records scaled values, which must be multiplied by the Multiplying Factor (MF) to obtain actual demand and energy values. This ensures correct billing quantities for both demand (kVA) and energy (kVAh).

$$\text{Billing kVAh} = \text{Recorded kWh} \times \text{MF} \div \text{PF}$$

These parameters ensure that demand charges, energy charges, and prepaid deductions for HT consumers are computed accurately and in full compliance with PED Mizoram and JERC-approved HT tariff principles.

b) Demand Charge Calculation

For HT Category-1 (Domestic Service), demand charges are calculated based on the Billing Demand, which is defined as the higher of the Contract Demand (kVA) or the Recorded Maximum Demand (MD) in kVA during the billing period. The demand charge represents the cost associated with maintaining system capacity to meet the consumer's demand and is applied as per the approved HT tariff.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. However, if at any time during the billing month the Recorded Maximum Demand exceeds the Contract Demand, the Billing Demand is revised to the higher Recorded MD. The prepaid engine continuously tracks the maximum recorded MD throughout the month, and reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD for the month is greater than the Contract Demand, the demand charge is recalculated on the basis of the maximum recorded MD, and the differential amount is adjusted against the prepaid balance on the last day of the billing cycle, with the current balance updated accordingly.

Demand Charge Rate: ₹50.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX (Contract Demand, Maximum Recorded MD)}$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge} \times (\text{Bill Days} / \text{Total Days in Month})$$

This ensures correct recovery of demand charges in prepaid mode and remains fully aligned with PED Mizoram and JERC-approved HT tariff principle

c) Energy Charge Calculation

For HT Category–I (Domestic Service), energy charges are calculated based on **apparent energy consumption (kVAh)** as per the approved HT tariff. Although the meter may record active energy in kWh, billing is carried out in kVAh to account for the impact of power factor on system loading.

In the prepaid system, interval-wise or daily energy consumption is received from the meter through ZONOS MDM. The recorded energy is first scaled using the **Multiplying Factor (MF)** and then converted from kWh to kVAh using the **Power Factor (PF)**. Energy charges are deducted from the prepaid balance progressively based on actual consumption.

Energy Charge Rate: ₹8.65 per kVAh

$$\text{Billing kVAh} = \text{Recorded kWh} \times \text{MF} \div \text{PF}$$

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

This ensures that energy charges in prepaid mode are computed accurately on apparent energy consumption and remain fully aligned with PED Mizoram and JERC-approved HT tariff principles.

d) Surcharge Calculation

As per PED Mizoram tariff provisions, a surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges for HT Category–I (Domestic Service). In the prepaid system, the surcharge is calculated after determining the applicable demand and energy charges and is deducted from the consumer's prepaid balance along with other charges.

The surcharge amount is computed on a provisional basis during the billing period and is finally reconciled on the last day of the billing cycle along with demand charge reconciliation, if any. The prepaid balance is updated accordingly after applying the surcharge.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

In the prepaid system for HT Category-I (Domestic Service), energy consumption is recorded on a daily or interval basis and priced using the applicable HT tariff logic. Energy charges are deducted progressively from the prepaid balance based on actual apparent energy consumption (kVAh). Demand charges are provisionally considered based on Contract Demand, and surcharge is computed on the applicable demand and energy charges.

Throughout the billing cycle, the prepaid engine maintains a running balance by deducting daily energy charges and provisional demand-related charges. On the last day of the billing cycle, a final reconciliation is carried out to account for the maximum recorded MD, applicable demand charges, and surcharge, if any. The differential amount arising from this reconciliation is adjusted against the prepaid balance, and the current balance is updated accordingly.

- Daily Energy Charge = Daily kVAh × Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand × Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) × 2%
- Daily Gross Deduction = Daily Energy Charge + Daily Provisional Demand Charge + Daily Provisional Surcharge
- Current Balance (Day n) = Previous Balance - Daily Gross Deduction

e) End-of-Month Reconciliation (HT – Demand Charge)

During the billing month, demand charges are provisionally deducted on a daily basis using the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is performed only on the last day of the billing cycle.

If the maximum recorded MD exceeds the Contract Demand, the final demand charge is recalculated based on the Billing Demand (higher of Contract Demand or maximum recorded MD). The reconciliation amount is computed as the difference between the final demand charge and the sum of provisional daily demand charge deductions already made during the month. This differential amount is adjusted

against the prepaid balance on the last day of the billing cycle, and the current balance is updated accordingly.

On the last day of the billing cycle, a final reconciliation is performed:

- Billing Demand is finalised as the higher of Contract Demand or maximum recorded MD
- Final Demand Charge is computed based on Billing Demand
- Final Surcharge is computed at 2% on final demand and total energy charges

The reconciliation amount is calculated as the difference between final charges and the sum of provisional daily deductions already made. This differential amount is adjusted against the prepaid balance, and the current balance is updated only at this stage.

- Final Demand Charge = Billing Demand × Demand Charge Rate × (Bill Days / Total Days in Month)
- Final Surcharge = (Final Demand Charge + Total Energy Charge) × 2%
- Sum of Provisional Demand Charges = Σ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge = Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge) - (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

f) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Domestic Consumer. It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.

- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



3.1Domestic(HT)

7.2 HT Category - 2 : Non-Profit Public Service

7.2.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage to non-profit institutions and public service establishments, such as government offices, educational institutions, hospitals, charitable institutions, religious places, and other similar entities. Supply under this category is strictly intended for public or community service purposes and excludes any commercial or profit-oriented activity.

7.2.2 Rates

Component	Rate
Demand Charge	₹60.00 per month per kVA of Billing Demand
Energy Charge	₹8.75 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.2.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Non-Profit Public Service consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique **Consumer ID** and **Device ID**. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totaldaysinmonth \div BillDays$
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

Billing kVAh = Recorded kWh * MF ÷ PF

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹60.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX (Contract Demand, Maximum Recorded MD)}$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Energy data received from the meter is scaled using MF and converted from kWh to kVAh using PF. Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹8.75 per kVAh

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption

- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh * Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand * Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) * 2%
- Daily Gross Deduction= Daily Energy Charge+ Daily Provisional Demand Charge+ Daily Provisional Surcharge
- Current Balance (Day n)= Previous Balance - Daily Gross Deduction

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand * Demand Charge Rate * (Bill Days / Total Days in Month)
- Final Surcharge= (Final Demand Charge + Total Energy Charge) * 2%
- Sum of Provisional Demand Charges= Σ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge= Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge) - (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Non-Profit Public Service. It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final

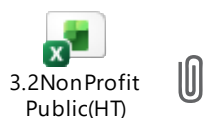
reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



7.3 HT Category - 3: Commercial Service

7.3.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage to premises used for commercial and business activities, such as large offices, hotels, malls, commercial complexes, private institutions, service establishments, and other profit-oriented activities. Supply under this category excludes domestic, non-profit public service, and industrial usage.

7.3.2 Rates

Component	Rate
Demand Charge	₹80.00 per month per kVA of Billing Demand
Energy Charge	₹8.90 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.3.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

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 Approved: Pradeep Narayana
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For HT Commercial Service consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = *Lastdayofcurrentmonth – InstallationDate*
- Bill Factor = *Totaldaysinmonth ÷ BillDays*
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$\text{Billing kVAh} = \text{Recorded kWh} \times \text{MF} \div \text{PF}$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹80.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX}(\text{Contract Demand, Maximum Recorded MD})$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Energy data received from the meter is scaled using MF and converted from kWh to kVAh using PF. Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹8.90 per kVAh

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with

daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh × Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand × Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) × 2%
- Daily Gross Deduction = Daily Energy Charge + Daily Provisional Demand Charge + Daily Provisional Surcharge
- Current Balance (Day n) = Previous Balance - Daily Gross Deduction

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand × Demand Charge Rate × (Bill Days / Total Days in Month)
- Final Surcharge = (Final Demand Charge + Total Energy Charge) × 2%
- Sum of Provisional Demand Charges = $\sum$ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge = $\sum$ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge) - (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)

- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Commercial Service. It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



3.3Commercial(HT)

7.4 HT Category - 4: Public Water Works (PWW)

7.4.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage for public water supply systems, including water pumping stations, treatment plants, and allied infrastructure operated by government departments, public authorities, municipal bodies, or other authorised agencies. Supply under this category is restricted to activities related to public water supply and excludes commercial or industrial usage.

7.4.2 Rates

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 Version: 01.00
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Component	Rate
Demand Charge	₹90.00 per month per kVA of Billing Demand
Energy Charge	₹9.85 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.4.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Public Water Works (PWW) consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = *Lastdayofcurrentmonth – InstallationDate*
- Bill Factor = *Totaldaysinmonth ÷ BillDays*
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$\text{Billing kVAh} = \text{Recorded kWh} \times \text{MF} \div \text{PF}$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹90.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX}(\text{Contract Demand, Maximum Recorded MD})$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Energy data received from the meter is scaled using the Multiplying Factor (MF) and converted from kWh to kVAh using the Power Factor (PF). Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹9.85 per kVAh

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh × Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand × Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) × 2%
- Daily Gross Deduction = Daily Energy Charge + Daily Provisional Demand Charge + Daily Provisional Surcharge
- Current Balance (Day n) = Previous Balance - Daily Gross Deduction

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand × Demand Charge Rate × (Bill Days / Total Days in Month)
- Final Surcharge= (Final Demand Charge + Total Energy Charge) × 2%
- Sum of Provisional Demand Charges= Σ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge= Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge) - (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Public Water Works (PWW). It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation


[3.4PublicWater\(HT\)](#)


7.5 HT Category - 5: Irrigation & Agriculture Service

7.5.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage for irrigation and agricultural purposes, including pumping for agriculture, horticulture, and allied farming activities. Supply under this category is restricted exclusively to agricultural use and excludes commercial, industrial, or domestic applications.

7.5.2 Rates

Component	Rate
Demand Charge	₹50.00 per month per kVA of Billing Demand
Energy Charge	₹3.85 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.5.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Irrigation & Agriculture Service consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totaldaysinmonth \div BillDays$
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$Billing\ kVAh = Recorded\ kWh \times MF \div PF$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹50.00 per kVA per month

Billing Demand (kVA) = MAX(Contract Demand, Maximum Recorded MD)

Demand Charge = Billing Demand * Demand Charge Rate * (Bill Days / Total Days in Month)

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Meter-recorded energy is scaled using the Multiplying Factor (MF) and converted from kWh to kVAh using the Power Factor (PF). Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹3.85 per kVAh

Energy Charge = Billing kVAh * Energy Charge Rate

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

Surcharge = (Demand Charge + Energy Charge) * 2%

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh * Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand * Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) * 2%
- Daily Gross Deduction= Daily Energy Charge+ Daily Provisional Demand Charge+ Daily Provisional Surcharge
- Current Balance (Day n)= Previous Balance - Daily Gross Deduction

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand * Demand Charge Rate * (Bill Days / Total Days in Month)
- Final Surcharge= (Final Demand Charge + Total Energy Charge) * 2%
- Sum of Provisional Demand Charges= Σ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge= Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge) - (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Irrigation & Agriculture Service. It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



3.5Irrigation(HT)

7.6 HT Category – 6: Industrial Service (Option-1)

7.6.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage to industrial consumers, including manufacturing units, processing plants, factories, workshops, and other industrial establishments. This option applies where billing is carried out without Time-of-Day (TOD) differentiation and energy charges are levied at a uniform rate for all units consumed.

7.6.2 Rates

Component	Rate
Demand Charge	₹80.00 per month per kVA of Billing Demand
Energy Charge	₹9.05 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.6.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Industrial Service (Option-1) consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totaldaysinmonth \div BillDays$
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$Billing\ kVAh = Recorded\ kWh \times MF \div PF$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹80.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX}(\text{Contract Demand, Maximum Recorded MD})$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Energy data received from the HT meter is scaled using the Multiplying Factor (MF) and converted from kWh to kVAh using the Power Factor (PF). Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹9.05 per kVAh

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh × Energy Charge Rate

- Daily Provisional Demand Charge = $(\text{Contract Demand} \times \text{Demand Charge Rate}) \div \text{Total Days in Month}$
- Daily Provisional Surcharge = $(\text{Daily Energy Charge} + \text{Daily Provisional Demand Charge}) \times 2\%$
- Daily Gross Deduction = $\text{Daily Energy Charge} + \text{Daily Provisional Demand Charge} + \text{Daily Provisional Surcharge}$
- Current Balance (Day n) = $\text{Previous Balance} - \text{Daily Gross Deduction}$

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = $\text{Billing Demand} \times \text{Demand Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$
- Final Surcharge = $(\text{Final Demand Charge} + \text{Total Energy Charge}) \times 2\%$
- Sum of Provisional Demand Charges = $\sum (\text{Daily Provisional Demand Charge})$
- Sum of Provisional Surcharge = $\sum (\text{Daily Provisional Surcharge})$
- Reconciliation Amount = $(\text{Final Demand Charge} + \text{Final Surcharge}) - (\text{Sum of Provisional Demand Charges} + \text{Sum of Provisional Surcharge})$
- Updated Current Balance = $\text{Pre-Reconciliation Balance} - \text{Reconciliation Amount}$

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Industrial Service (Option-1). It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

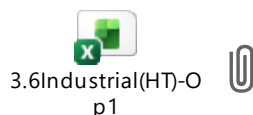
Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



7.7 HT Category - 6: Industrial Service (Option-2 – TOD Tariff)

7.7.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage to industrial consumers opting for Time-of-Day (TOD) based tariff. It applies to manufacturing units, factories, processing plants, and other industrial establishments where energy consumption varies significantly across different time blocks of the day.

7.7.2 Rates

Component	Rate
Energy Charge – Normal TOD (06:00–17:00 Hrs)	₹10.05 per kVAh
Energy Charge – Peak TOD (17:00–22:00 Hrs)	₹11.55 per kVAh
Energy Charge – Off-Peak TOD (22:00–06:00 Hrs)	₹8.55 per kVAh
Demand Charge	₹80.00 per month per kVA of Billing Demand
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.7.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Industrial Service (Option-2 – TOD Tariff) consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totaldaysinmonth \div BillDays$
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$\text{Billing kVAh} = \text{Recorded kWh} \times \text{MF} \div \text{PF}$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹80.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX}(\text{Contract Demand, Maximum Recorded MD})$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation (TOD-wise)

Energy charges are calculated based on apparent energy consumption (kVAh) segregated by Time-of-Day (TOD) slots. The HT meter records TOD-wise kVAh, which is scaled using MF and converted using PF before billing.

TOD Time Blocks:

- Normal TOD: 06:00 hrs to 17:00 hrs
- Peak TOD: 17:00 hrs to 22:00 hrs
- Off-Peak TOD: 22:00 hrs to 06:00 hrs

$$\text{Normal TOD Energy Charge} = \text{Normal TOD kVAh} \times 10.05$$

$$\text{Peak TOD Energy Charge} = \text{Peak TOD kVAh} \times 11.55$$

$$\text{Off-Peak TOD Energy Charge} = \text{Off-Peak TOD kVAh} \times 8.55$$

$$\text{Total Energy Charge} = \text{Normal TOD} + \text{Peak TOD} + \text{Off-Peak TOD}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily or interval-wise provisional deductions:

- TOD-wise energy charges based on actual kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2%

$$\text{Daily Energy Charge} = (\text{Normal TOD kVAh} \times 10.05) + (\text{Peak TOD kVAh} \times 11.55) + (\text{Off-Peak TOD kVAh} \times 8.55)$$

$$\text{Daily Provisional Demand Charge} = (\text{Contract Demand} \times \text{Demand Charge Rate}) \div \text{Total Days in Month}$$

$$\text{Daily Provisional Surcharge} = (\text{Daily Energy Charge} + \text{Daily Provisional Demand Charge}) \times 2\%$$

$$\text{Daily Gross Deduction} = \text{Daily Energy Charge} + \text{Daily Provisional Demand Charge} + \text{Daily Provisional Surcharge}$$

$$\text{Current Balance (Day } n) = \text{Previous Balance} - \text{Daily Gross Deduction}$$

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand $\times$ Demand Charge Rate $\times$ (Bill Days / Total Days in Month)
- Final Surcharge = (Final Demand Charge + Total Energy Charge) $\times$ 2%
- Sum of Provisional Demand Charges = Σ (Daily Provisional Demand Charge)

- Sum of Provisional Surcharge= Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge)- (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Industrial Service (Option-2 – TOD Tariff) .It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Normal TOD Units	Peak TOD Units	Off-Peak TOD Units	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1								
Day 2								
Day 3								
Day 4								
Day 5								

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



3.6Industrial(HT)-O
p2_TOD


7.8 HT Category – 7: Bulk Supply within the State

7.8.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage for bulk supply within the State, including supply to bulk consumers such as government utilities, large public entities, distribution licensees, or other authorised agencies drawing power in bulk for onward distribution or specified large-scale use. Supply under this category is governed strictly by approved bulk supply arrangements.

7.8.2 Rates

Component	Rate
Demand Charge	₹90.00 per month per kVA of Billing Demand
Energy Charge	₹6.85 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.8.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Bulk Supply within the State consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totaldaysinmonth \div BillDays$
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$Billing\ kVAh = Recorded\ kWh \times MF \div PF$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹90.00 per kVA per month

$$Billing\ Demand\ (kVA) = MAX(Contract\ Demand, Maximum\ Recorded\ MD)$$

$$Demand\ Charge = Billing\ Demand \times Demand\ Charge\ Rate \times (Bill\ Days / Total\ Days\ in\ Month)$$

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Meter-recorded energy is scaled using the Multiplying Factor (MF) and converted from kWh to kVAh using the Power Factor (PF). Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹6.85 per kVAh

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh × Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand × Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) × 2%
- Daily Gross Deduction = Daily Energy Charge + Daily Provisional Demand Charge + Daily Provisional Surcharge
- Current Balance (Day n) = Previous Balance - Daily Gross Deduction

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand × Demand Charge Rate × (Bill Days / Total Days in Month)
- Final Surcharge= (Final Demand Charge + Total Energy Charge) × 2%
- Sum of Provisional Demand Charges= Σ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge= Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge)- (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Bulk Supply within the State.It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kvAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.
- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation




3.7BulkSupply(HT)

7.9 HT Category – 8: Electric Vehicle Charging Stations

7.9.1 Applicability

This category is applicable for electricity supply at High Tension (HT) voltage for electric vehicle (EV) charging stations, including public and private charging infrastructure operated by government agencies, utilities, commercial entities, or authorised service providers. Supply under this category is restricted exclusively to EV charging activities and excludes any non-EV or industrial usage.

7.9.2 Rates

Component	Rate
Demand Charge	₹75.00 per month per kVA of Billing Demand
Energy Charge	₹8.65 per kVAh (for all units)
Billing Demand	Higher of Contract Demand (kVA) or Recorded Maximum Demand (kVA)
Surcharge	2% on (Demand Charge + Energy Charge)

7.9.3 Prepayment Configuration

a) Master Data Sync and Initial Configuration

For HT Electric Vehicle Charging Stations consumers, the prepaid engine is integrated with the ZONOS Meter Data Management (MDM) system. Consumer, meter, and electrical parameters are synchronised using a unique Consumer ID and Device ID. The following parameters are received and validated for prepaid tariff configuration:

- Contract Demand (kVA)
- Installation Date (dd-mm-yyyy)
- Bill Days = $Lastdayofcurrentmonth - InstallationDate$
- Bill Factor = $Totaldaysinmonth \div BillDays$
- Power Factor (PF)
- Multiplying Factor (MF)
- Opening Balance (₹)

$$Billing\ kVAh = Recorded\ kWh \times MF \div PF$$

b) Demand Charge Calculation

Demand charges are calculated based on the Billing Demand, defined as the higher of the Contract Demand or the maximum recorded MD during the billing cycle.

In the prepaid system, demand charges are provisionally deducted based on the Contract Demand. The prepaid engine continuously tracks the maximum recorded MD throughout the month. Reconciliation is carried out only on the last day of the billing cycle. If the maximum recorded MD exceeds the Contract Demand, the Billing Demand is revised accordingly, and the differential demand charge is adjusted

against the prepaid balance, with the current balance updated on the last day of the billing cycle.

Demand Charge Rate: ₹75.00 per kVA per month

$$\text{Billing Demand (kVA)} = \text{MAX}(\text{Contract Demand, Maximum Recorded MD})$$

$$\text{Demand Charge} = \text{Billing Demand} \times \text{Demand Charge Rate} \times (\text{Bill Days} / \text{Total Days in Month})$$

c) Energy Charge Calculation

Energy charges are calculated on apparent energy consumption (kVAh). Meter-recorded energy is scaled using the Multiplying Factor (MF) and converted from kWh to kVAh using the Power Factor (PF). Energy charges are deducted progressively from the prepaid balance based on actual consumption.

Energy Charge Rate: ₹8.65 per kVAh

$$\text{Energy Charge} = \text{Billing kVAh} \times \text{Energy Charge Rate}$$

d) Surcharge Calculation

A surcharge at the rate of 2% is applicable on the total of Demand Charges and Energy Charges. In prepaid mode, surcharge is provisionally deducted along with daily energy and demand charges and is finally reconciled on the last day of the billing cycle.

Surcharge Rate: 2%

$$\text{Surcharge} = (\text{Demand Charge} + \text{Energy Charge}) \times 2\%$$

e) Daily Prepaid Deduction and Balance Update

During the billing month, the prepaid engine performs daily (or interval-wise) provisional deductions to maintain real-time balance visibility:

- Energy charges based on actual daily kVAh consumption
- Provisional demand charges based on Contract Demand
- Provisional surcharge at 2% on daily energy and provisional demand charges

- Daily Energy Charge = Daily kVAh × Energy Charge Rate
- Daily Provisional Demand Charge = (Contract Demand × Demand Charge Rate) ÷ Total Days in Month
- Daily Provisional Surcharge = (Daily Energy Charge + Daily Provisional Demand Charge) × 2%
- Daily Gross Deduction = Daily Energy Charge + Daily Provisional Demand Charge + Daily Provisional Surcharge
- Current Balance (Day n) = Previous Balance - Daily Gross Deduction

f) End-of-Month Reconciliation (Demand + Surcharge)

On the last day of the billing cycle, final reconciliation is carried out as per the standard HT reconciliation clause:

- Billing Demand finalised as higher of Contract Demand or maximum recorded MD
- Final Demand Charge recalculated
- Final Surcharge recalculated at 2% on final demand and total energy charges

The reconciliation amount is computed as the difference between final charges and the sum of provisional daily demand and surcharge deductions, and the prepaid balance is updated accordingly.

- Final Demand Charge = Billing Demand × Demand Charge Rate × (Bill Days / Total Days in Month)
- Final Surcharge = (Final Demand Charge + Total Energy Charge) × 2%
- Sum of Provisional Demand Charges = Σ (Daily Provisional Demand Charge)
- Sum of Provisional Surcharge = Σ (Daily Provisional Surcharge)
- Reconciliation Amount = (Final Demand Charge + Final Surcharge) - (Sum of Provisional Demand Charges + Sum of Provisional Surcharge)
- Updated Current Balance = Pre-Reconciliation Balance - Reconciliation Amount

g) Daily Prepaid Deduction View:

The table below provides an indicative view of daily prepaid deductions for an HT Electric Vehicle Charging Stations. It shows how energy charges, provisional demand charges, and surcharge are deducted on a daily basis from the prepaid balance. Final reconciliation of demand charges and surcharge, if any, is carried out only on the last day of the billing cycle, as described in the reconciliation section.

Day	Units Consumed (kVAh)	Demand Charge (₹)	Energy Charges (₹)	Surcharge (₹)	Gross Deduction (₹)	Current Balance (₹)
Day 1						
Day 2						
Day 3						
Day 4						
Day 5						

Note:

- Energy charge is calculated on actual daily kVAh consumption.
- Demand charge is provisionally deducted daily based on Contract Demand.
- Surcharge is calculated daily at 2% on energy + provisional demand charges.
- No reconciliation occurs mid-month.

- Any adjustment due to maximum recorded MD exceeding Contract Demand, along with surcharge impact, is applied only on the last day of the billing cycle, and the current balance is updated accordingly.

g) Link to the Sample Calculation



8. Acceptance

Satnam Global with MDM Partner Cuculus India has to configure the LT and HT tariff structures in the prepayment engine strictly in line with the approved PED Mizoram tariff orders and JERC regulatory principles. The configuration covers applicability, rate structures, daily prepaid deductions, surcharge application, and end-of-month reconciliation, ensuring transparency, accuracy, and regulatory compliance across all consumer categories.

We hereby submit this documentation for review and approval by PED Mizoram, and seek confirmation that the tariff logic, calculation methodology, and prepaid implementation are acceptable for operational rollout.

Upon approval, the same configuration will be treated as the final baseline for prepaid deployment across applicable LT and HT consumer categories.